

Computer Networks - Chapter 5 Notes

Transport Layer

1. Introduction to Transport Layer:

- Provides end-to-end communication between processes.
- Ensures reliable delivery, sequencing, error control, and flow control.

2. Elements of Transport Protocol:

- Addressing (Port numbers: source & destination).
- Multiplexing & Demultiplexing.
- Error Detection & Correction.
- Flow Control (Sliding Window).
- Connection Establishment & Termination.

3. Congestion Control Mechanisms:

- Purpose: Prevent overload in the network.
- Leaky Bucket Algorithm.
- Token Bucket Algorithm.
- TCP: Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery.

4. Transmission Control Protocol (TCP):

- Connection-oriented, reliable, byte-stream protocol.
- 3-Way Handshake for connection setup.
- Provides error recovery, sequencing, flow & congestion control.
- Uses acknowledgments and retransmissions.

5. User Datagram Protocol (UDP):

- Connectionless, unreliable protocol.
- Provides low-latency communication (VoIP, gaming).
- No congestion control, faster but less reliable.

6. Remote Procedure Call (RPC):

- Allows program on one computer to execute procedures on another computer.
- Hides details of network communication.
- Implementation Semantics:
 - * Exactly once, At most once, At least once.

7. BSD Sockets:

- Programming API for network communication.
- Used to create client-server applications.
- Steps: socket(), bind(), listen(), accept(), connect(), send(), recv().

8. Client-Server Applications:

- Client sends request, server provides service.
- Examples: Web Browsing (HTTP), Email (SMTP/IMAP).

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